Arista Access Node YEL-ACC-32 QoS Congestion Event Resolution SOP

Node: YEL-ACC-32 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_7694

Description

The described failure domain, specifically QoS / Congestion Events (Microbursts, interface tail drops, QoS queue starvation), on the Arista Access Node YEL-ACC-32 at the Yelahanka Sub-station is attributed to the inherent complexities of Quality of Service (QoS) policing and shaping within the network. The Access Node, operating within the confines of the Arista EOS CLI dialect, relies heavily on the Traffic Management and QoS features to ensure prioritization of critical traffic streams. However, when the configuration settings are inadequately fine-tuned or encounter unforeseen congestion, the system may succumb to microbursts, interface tail drops, or QoS queue starvation, precipitating a Critical Alarm Code ALARM_7694. This document outlines a standardized resolution procedure to rectify the aforementioned QoS Congestion Event by meticulously addressing the potential root causes, executing diagnostic commands to pinpoint the issue, and implementing corrective measures to restore normal network functionality.

Pre-requisite Checks

- check: Verify the power and environmental status of the YEL-ACC-32 node and the Yelahanka Sub-station., description: Ensure that the node and the sub-station are powered on and within acceptable environmental limits.

- check: Confirm the status of redundant links and connectivity between the node and adjacent network elements., description: Verify that the redundant links are operational and free from any configuration-related issues.

- check: Review the system logs for any recent configuration changes or error-related events., description: Analyze the system logs to determine if any recent configuration modifications or error incidents may be contributing to the QoS Congestion Event.

Potential Root Causes

Cause	Probability
Inadequate QoS Configuration	High
Buffer Overflow or Interface Congestion	Medium
Hardware or Software Failure	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware profile detail, Expected Output: Hardware
Profile Summary: ..., Description: This command helps in identifying
potential hardware-related issues, such as buffer overflows or
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```
interface congestion, that may be contributing to the QoS Congestion
Event.

Command: show qos statistics interface all, Expected Output: Interface
QoS Statistics: ..., Description: This command provides detailed QoS
statistics for all interfaces, including packet drops, tail drops, and
queue drops, which can help in pinpointing the root cause of the
congestion event.

Command: show qos configuration, Expected Output: QoS Configuration:
..., Description: This command displays the current QoS configuration,
including policing and shaping settings, which can help in identifying
any configuration-related issues contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Adjust QoS Configuration, Command: configure terminal qos policing qos shaping , Description: Modify the QoS configuration to optimize policing and shaping settings based on the diagnostic findings. Ensure that the policing and shaping rates are aligned with the actual traffic requirements to prevent congestion.

Step 2: Step: Clear Buffer Overflows or Interface Congestion, Command: clear hardware buffers , Description: Clear any buffer overflows or interface congestion issues identified during the diagnostic phase. This step ensures that the buffers and interfaces are reset to their default state, preventing any potential congestion.

Step 3: Step: Verify System Logs, Command: show log, Description: Review the system logs to ensure that the QoS Congestion Event has been resolved and that there are no further errors or warnings related to QoS or congestion.

Post-Check Verification

```
Command: show qos statistics interface all, Expected Output: Interface
QoS Statistics: ..., Description: Verify that the QoS statistics have
returned to normal, indicating that the congestion event has been
resolved.

Command: show qos configuration, Expected Output: QoS Configuration:
..., Description: Verify that the QoS configuration has been
successfully modified to prevent future congestion events.
```

Roll-back Procedures

- Step: Restore Original QoS Configuration, Command: configure terminal qos policing , Description: Restore the original QoS configuration to its previous state in case the resolution steps fail to resolve the congestion event.

- Step: Clear Buffer Overflows or Interface Congestion (Again), Command: clear hardware buffers , Description: Clear any buffer overflows or interface congestion issues that may have occurred during the resolution phase.